Confirming Process, Missing Function: The Uninstrumented Oversight Layer in AI Transparency and Control Research

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Abstract

Audit infrastructure for AI-assisted workflows can document that human review occurred without establishing whether substantive evaluative judgment was exercised. The two system states most consequential for safety, oversight that genuinely examined an output and oversight that did not, can be indistinguishable in the observable record. Logs and approval chains document that review occurred and that required parties participated, but neither speaks to whether the cognitive function that gives human oversight its safety value was actually exercised. This invisibility is not an incidental measurement failure; it reflects a structural gap in what current AI transparency and governance infrastructure can observe.

Observational research in two deployment contexts identifies a recurring degradation pattern: oversight quality shifts from substantive engagement to procedural confirmation while the governance record remains formally compliant throughout. Current detection, interpretability, and governance methods do not distinguish these states because they instrument the model and its outputs rather than the human oversight function. This paper argues that the oversight function itself must be treated as a system component requiring observation and characterization, identifies the structural conditions that drive its degradation, and proposes a research direction focused on instrumenting the oversight layer through structural indicators including correction-rate trajectory, validation-load ratio, and dependency accumulation. The contribution is a measurement and design agenda for the governance infrastructure that current frameworks lack.

Keywords

human oversight, AI transparency, LLM-assisted knowledge work, governance instrumentation, human-in-the-loop

1 Introduction

The governance record of an AI-assisted workflow can be procedurally complete while remaining uninformative about the most consequential variable in the system: whether the human reviewer who approved an output was substantively engaged with it or was performing procedural confirmation. Although logs document that review occurred and approval chains confirm that required parties participated, neither reveals whether the cognitive function that gives human oversight its safety value was actually exercised. This is the fundamental structural flaw in human-in-the-loop oversight as commonly implemented and audited: the gap between procedural record and functional reality is not an incidental measurement limitation but a structural consequence of governance infrastructure that instruments the model and its outputs rather than the oversight function on which its safety properties depend.

Governance requirements that mandate human oversight of AI-generated outputs have proliferated across regulatory frameworks, organizational policies, and safety standards. The EU AI Act's Article 14 imposes human oversight requirements on all high-risk AI systems, requiring providers to design systems that natural persons can effectively oversee [11]. The NIST AI Risk Management Framework addresses human oversight in its Govern function, particularly Govern 3.2 and Appendix C, recommending documented accountability for AI deployment decisions [47]. In the financial sector, model risk management guidance issued by the Federal Reserve and OCC as SR 11-7 served for over a decade as the baseline framework that institutions extended to cover AI-assisted decision tools [5]; that guidance was rescinded in April 2026 and replaced by a more explicitly risk-based, principles-driven framework [6], but the structural pattern of extending established financial-sector model governance to AI-assisted tools persists in the successor framework, and similar adaptations are underway in clinical and legal practice contexts. Yet the functional quality of that human presence is rarely instrumented with anything beyond the procedural record itself, and the conditions under which it degrades are poorly understood. Existing methods for detecting AI agent behavior, ensuring model interpretability, and governing AI deployments share a tacit assumption: that the oversight function on which their safety properties depend is sufficiently stable that it need not itself be observed or maintained. We argue that this assumption is not reliably warranted, and that when it fails, the failure is invisible to the infrastructure designed to verify it.

The gap is not a claim that no scholarship studies oversight quality. Work on the responsibility-gap framework developed for autonomous weapons systems [39], the conditions under which human reviewers appropriately rely on or override AI recommendations [41], automation complacency [30, 31], the functional limits of oversight policies [17], and the challenges of testing for compliance with human-oversight requirements in AI regulation [22] has treated the human reviewer as a consequential and degradable system component. What remains incomplete is the systematic integration of these findings into the transparency, audit, and governance mechanisms that are actually used to certify AI deployments. Those mechanisms predominantly instrument the model layer, leaving the oversight function comparatively underobserved. This paper extends the observational characterization of the centaur and cyborg architectural patterns [27], moving from descriptive characterization of the patterns to a prescriptive measurement and design agenda for the governance infrastructure that current frameworks lack. The argument that follows is not that human reviewers are incompetent or insufficient; it is that the operational conditions under which substantive oversight of high-capability AI systems is achievable are novel, only partially analogous to prior automation contexts, and not yet systematically characterized.

Beyond writing review, the same gap applies wherever natural-language instruction operates as the interface between human authority and automated execution. When the programming language becomes English, prompts that direct AI agents in high-criticality environments inherit the same observational identity problem as approval records: the instruction was issued, but whether the issuer engaged substantively with the implications is not separately recorded. The scope of the oversight problem this paper develops is therefore not limited to document review workflows but extends to any operational context in which natural-language interaction substitutes for the more constrained interfaces that earlier generations of automation made available.

We develop this argument in four steps. We first situate the oversight function within a supervisory control framework, establishing why it requires instrumentation rather than assumption. We then present observational evidence of a predictable degradation pattern in which oversight persists structurally while its functional role erodes, and in which the degraded and functional states are observationally identical from the vantage points available to current frameworks. We then analyze why each of the three primary transparency functions, detection, interpretability, and governance, is insufficient to close this gap. We close with a research agenda for extending transparency and control work to the oversight layer itself, positioned at the intersection of research inquiry and practical implementation so that the proposed instrumentation can be built, deployed, and tested under operational conditions rather than only investigated in laboratory settings.

2 The Oversight Function as a System Component

Across regulatory frameworks, organizational policies, and safety standards, human oversight is widely prescribed but rarely defined with precision. What these prescriptions share is a treatment of oversight as a binary condition: either present or absent. This framing is insufficient for safety analysis because it treats the oversight function as a switch rather than as a system component with its own operating conditions, degradation modes, and failure signatures.

Supervisory control theory offers a more useful starting point. Within this framework, human operators do not directly execute controlled processes; they monitor automated subsystems, set parameters, intervene on exception, retain authority over consequential decisions, and audit system behavior across operational periods [32]. The human role is structurally essential to the system's safety properties rather than incidental to them. Critically, the theory treats the operator's functional state as a variable requiring active management, not a stable background condition. Workload, vigilance, calibration, and situational awareness are all recognized as components of the oversight function that can be measured, degraded, and designed for.

The supervisory control framing is not confined to the industrial and aviation contexts in which it was developed. Modern IT operations provide a direct structural parallel. Site reliability engineers do not directly execute the systems they manage; they monitor automated infrastructure, respond to alerting signals, intervene on exception, and retain authority over escalation and remediation decisions. Platform engineering operates at higher scale: orchestration design patterns deploy and manage automated agents across thousands of hosts, with supervisory control exercised through declarative policy, configuration management, and rollback authority rather than through direct execution. On-call workflows, runbook execution, incident response, and orchestration policy enforcement are all supervisory functions over automated systems, and the degradation patterns documented in automation research appear in recognizable form in these contexts. Alert fatigue in security operations centers and clinical decision support, and monitoring lapses in highly automated aviation cockpits, all describe the same structural failure mode under sustained operational conditions [1, 40, 48]. LLM-assisted knowledge work is the most recent context in which this pattern has emerged, not the first.

Applying this framework to oversight of AI systems produces a direct implication: the oversight function should be treated as a system component in its own right, with observable state, measurable quality, and failure modes that can be characterized and addressed through design. This is not a claim that engaged oversight is by itself sufficient for high-stakes AI deployment. Structural limits apply to engaged reviewers as well as procedural ones, and those limits remain regardless of how engaged the reviewer is. The narrower claim made here is that distinguishing the engaged state from the procedural one is the precondition for any analysis of when engaged oversight is sufficient.

The automation complacency literature has established that the variation in oversight quality is not incidental. Automation bias, the tendency of decision makers to substitute automated cues for vigilant information seeking and processing, produces errors of both omission and commission that increase as automated systems become more reliable and operators more accustomed to trusting them [29, 30]. A complementary line of work shows that automating routine cognitive tasks erodes the attentional and skill-based conditions on which competent oversight depends [2]. Recent work draws the connection forward into generative AI directly: Simkute and colleagues map four ironies of GenAI productivity loss (the production-to-evaluation shift, workflow restructuring, task interruptions, and task-complexity polarization) onto the older automation literature, situating the current generation of usability and oversight failures as recognizable expressions of patterns documented for over three decades [44]. The pattern is well-established. What is qualitatively new under LLMs is the suppression of the surface signals on which detection has historically depended.

LLM-mediated knowledge work introduces a specific intensifier that distinguishes the current situation from earlier automation contexts. In prior automation settings, the boundary between correct and incorrect automated output was often legible: a sensor reading exceeded a threshold, a calculation produced a visible anomaly, a process deviated from expected parameters. High-capability language models remove many of these signals. Their outputs are fluent, internally coherent, and often difficult to distinguish from competent human-authored text under ordinary review conditions, especially for non-expert, time-pressured, or cross-domain reviewers, whether or not those outputs are correct. This is a qualitative difference from prior automation contexts: numeric and process-based automation produced outputs with legible error signatures, whereas language model outputs suppress the surface features that ordinarily trigger scrutiny regardless of underlying correctness. An aircraft sensor reading that exceeds threshold produces a discrete alert that orients the operator to a specific anomaly, whereas a fluent LLM paragraph containing a fabricated citation, a misattributed quotation, or an inverted causal claim produces no equivalent alert; the error is present but the surface form does not announce it. This applies across the range of errors language models produce. Subtle errors are difficult to detect under any conditions; what is distinctive about LLM-mediated knowledge work is that exuberantly wrong outputs, those that are confidently expressed but transparently incorrect to a domain expert, are also routinely missed under ordinary review conditions, because the same rhetorical fluency that suppresses scrutiny of subtle errors normalizes the confidence of dramatic ones. Survey evidence from knowledge workers indicates that higher confidence in GenAI output is associated with reduced self-reported critical thinking effort [24].

Randazzo and colleagues characterize the behavioral level at which this artifact-proxy gap manifests, identifying three distinct modes by which knowledge workers integrate generative AI into their workflows: cyborg, centaur, and self-automator [36]. The architectural framing developed here is complementary to that behavioral one. Architectures constrain the workflow conditions under which different co-creation modes are sustainable, while the modes themselves operate at the level of how individuals work within those conditions. Controlled studies of reliance interventions in LLM-assisted decision tasks find that common approaches reduce overreliance only partially, do not reliably produce appropriate reliance, and in some conditions increase user confidence even following incorrect reliance decisions [4].

The argument developed here is not that AI assistance invariably degrades oversight quality. Under appropriate conditions, AI systems can improve reviewer calibration, consistency, and coverage. A well-designed explanation interface that surfaces the model's uncertainty alongside its output gives reviewers a legible signal for allocating scrutiny. A system that flags outputs falling outside the distribution of previously validated cases directs reviewer attention toward precisely the outputs that warrant it. A workflow that structures the review task, decomposing complex outputs into verifiable components, can reduce the cognitive load that makes procedural confirmation a rational response to unsustainable demand. These are conditions under which AI assistance and substantive human oversight may be mutually reinforcing rather than in tension, though the precise constraints on reviewer capacity and the architectural competence of AI systems supporting complex review remain insufficiently characterized in the existing literature.

Whether these conditions actually obtain depends on the sociotechnical architecture in which an AI system is deployed, not on the model or the reviewer in isolation. This is not a soft claim about context-sensitivity. It is a structural claim about where causal responsibility for oversight quality lies: not in the model, not in the reviewer in isolation, but in the architecture that connects them, including workflow design, interface conditions, reviewer capacity allocation, incentive structures, and the audit mechanisms that record what happens. Architectures designed without explicit attention to these conditions will produce degraded oversight even when the model is capable and the reviewer is competent, because the conditions that encourage substantive and sustainable engagement have not been designed in.

Early observational patterns, not yet constituting empirical evidence, suggest that in the absence of deliberate design, some communities and deployment contexts exhibit responses more severe than procedural confirmation alone, including patterns consistent with what Shaw and Nave term cognitive surrender: the uncritical adoption of AI outputs with minimal scrutiny, in which the locus of evaluative judgment shifts from the person to the model rather than being supplemented by it [43]. These patterns appear to vary across communities and professional cultures in ways that are not yet well characterized. The conditions under which professional identity norms either resist or accelerate these responses represent a convergence point between the incentive-structural literature and research on how organizational and community contexts shape implicit cognitive responses to external pressure [9]. That convergence is a priority direction for the empirical program this paper opens.

The consequence is that in LLM-assisted knowledge work, model capability and oversight fragility may scale together rather than in opposition. A more capable model produces more fluent output, which suppresses scrutiny signals more effectively, which reduces the probability that the oversight function operates at full engagement, which reduces the value of the oversight that governance records certify as having occurred. This dynamic is structural rather than behavioral. It does not require individual reviewers to be negligent, undertrained, or poorly motivated. It requires only that the oversight function is treated as a stable substrate rather than as a component requiring care and feeding.

3 The Degradation Pattern

The claim that oversight degrades under operational conditions is not new. What distinguishes the pattern we describe from prior accounts of automation complacency and monitoring failure is a specific structural property: the degraded state and the functional state are observationally identical from the vantage points available to current governance infrastructure. What is being claimed here concerns the detectability of degradation rather than its frequency or severity.

Earlier observational research examined human-LLM collaboration across two deployment contexts, an academic knowledge-work setting observed over seven consecutive semesters and an enterprise setting observed over an extended operational period [27, 28]. Both contexts exhibited a three-phase degradation trajectory that emerged independently of governance approach. The two organizations held opposing governance postures, one permissive and one restrictive, yet produced convergent degradation trajectories. Observations from two contrasting deployment contexts are consistent with the interpretation that the degradation pattern is not reducible to policy posture alone, and that governance intervention at the policy level is unlikely to arrest it without architectural change.

Phase 1: Substantive Engagement. In the initial deployment period, the validation checkpoint operates as a genuine quality gate.

Reviewers examine AI-generated outputs against domain knowledge, identify errors at rates consistent with baseline output quality, and exercise authority over what proceeds downstream and what is corrected, rejected, or escalated. The observable signatures of this phase include elevated correction rates, modification patterns that target substantive content rather than surface features, and reviewer-initiated escalations when output quality falls below threshold. Phase 1 is sustained by conditions that match reviewer cognitive capacity to output volume and complexity, by calibration that is current against ground truth, and by incentive structures that reward validation quality alongside or above throughput.

Phase 2: Procedural Scanning. Under sustained operational conditions, a second phase emerges. The formal structure of the validation process remains intact: reviews occur, approvals are recorded, and governance artifacts are produced. The functional character of the review has shifted, however. Scrutiny has been replaced by pattern recognition. Reviewers identify outputs that match expected structure and confirm them; outputs that deviate from expected patterns receive closer attention. Correction rates decline not because output quality has improved but because the threshold for intervention has risen. The mechanism underlying this transition has a cognitive architecture. When the target attribute, whether an output has been substantively evaluated, is demanding or inaccessible, fast, automatic cognitive processing substitutes a more accessible heuristic: whether the review process has been completed. This is attribute substitution as described by Kahneman, in which a difficult judgment is replaced by an easier one the mind can answer more quickly [19]. This substitution is not deliberate; it reflects the motivated reasoning tendency to arrive at the conclusion that requires least cognitive effort [21]. Incentive structures that compensate on process completion rather than validation quality accelerate this substitution by making the heuristic attribute the measured and rewarded one, a phenomenon Choi and colleagues term surrogation [8]. The governance record does not reflect this transition. Documentation confirms that review occurred; it does not capture that the cognitive function underlying review has shifted from deliberate evaluation to automatic confirmation.

Phase 3: Formality. In the terminal phase, the validation checkpoint has become a procedural acknowledgment. Approval is recorded without substantive evaluation. The human role has shifted from controller to confirmer. The oversight function is operating at minimal engagement. Governance artifacts remain intact and complete. An external audit of the process would find all required steps documented, all approvals recorded, and all required parties involved. Nothing in the governance record distinguishes this state from Phase 1.

Two structural properties of this trajectory require emphasis. The first is that transition between phases does not involve a discrete failure event. Degradation accumulates within normal operational parameters, producing a baseline shift that event-driven detection mechanisms are not designed to identify, with no alarm trigger, no threshold crossing, and no incident to initiate retrospective examination of oversight quality. Personnel turnover compounds this invisibility: individuals who calibrated against Phase 1 standards rotate out, and the degraded state becomes the only documented operational reality for those who follow. The governance framework reflects designed intent; operational reality reflects accumulated drift that the framework cannot observe. The second is the observational identity problem. The two system states that matter most for safety analysis, a reviewer who examined an output, found it sound, and approved it, and a reviewer who did not examine the output and approved it procedurally, are identical from the perspective of the governance record. Both produce a logged review event, a recorded approval, and complete documentation. Neither leaves a trace in the audit infrastructure that distinguishes substantive engagement from procedural confirmation. Standard governance mechanisms confirm process integrity; they are not designed to observe functional integrity, and current infrastructure treats the two as equivalent when they are not.

This observational identity problem is not an incidental measurement gap. It is a structural consequence of where governance infrastructure is pointed. Audit mechanisms were designed to verify that the process was followed, not to observe the cognitive function within the process, and that design choice was reasonable when the risk was process non-compliance. It becomes a safety gap when the risk is process compliance coexisting with functional absence.

The FMEA analysis conducted in that earlier observational research assigns detection difficulty scores to the failure modes associated with this trajectory. Authority Drift and Blind Acceptance, the two failure modes most directly associated with Phase 3, receive among the highest detection difficulty scores in the analysis, reflecting precisely this observational identity problem. High-severity failure modes that are structurally difficult to detect represent the most consequential category of risk in any safety-critical system. The uninstrumented oversight layer is the condition that keeps these failure modes in that category [28].

4 Why Existing Transparency Functions Do Not Instrument the Oversight Layer

If the degradation is structurally invisible to current infrastructure, the question is why. Governance infrastructure designed to observe the model and its outputs continues to report normal operation when the oversight function degrades, because it is pointed at the wrong layer. The remainder of this section examines why each of the three primary transparency functions, detection, interpretability, and governance, is structurally unable to close the gap independently.

4.1 Detection

Methods for detecting AI agent behavior assume that the downstream human response to detected signals is itself reliable: anomaly detection surfaces unexpected model outputs, behavioral monitoring captures agent actions, and evaluation pipelines flag outputs that deviate from expected distributions. Each is designed to produce a signal that a human analyst will act on, but none instruments whether the analyst actually does.

The detection gap operates at two levels. The first is direct: no standard detection method monitors the functional state of the reviewer. There is no sensor pointed at whether the approval recorded in the governance log reflects substantive evaluation or procedural confirmation. The second is indirect and more consequential: LLM output fluency suppresses the signals that detection methods rely on to redirect human attention. A well-formed, internally coherent output that contains a consequential error does not trigger anomaly detection, does not surface as a behavioral deviation, and does not produce a distribution flag. It passes through the detection layer cleanly and arrives at the human reviewer without any signal that closer examination is warranted.

The most commonly proposed governance remedy for this fluency-suppression dynamic, AI writing identification, warrants explicit treatment because it would, if reliable, close the gap at the surface-feature layer. AI writing identification tools, like the plagiarism detectors that preceded them, are imperfect even at present and likely to become more so as model outputs improve and human-AI writing becomes increasingly hybrid. Recent empirical work documents the trajectory already in motion: while annotators who frequently use LLMs for writing tasks can detect current-generation outputs with high accuracy, annotators without that experience perform substantially worse, and detection reliability degrades further when adversarial humanization techniques are applied to outputs from newer reasoning models [38]. The reviewers actually positioned in operational governance contexts typically resemble the non-expert population in that study, not the LLM-fluent expert population. The window in which surface-feature detection meaningfully constrains the problem is closing, and even where detection is possible, the broader phenomenon of human trust calibration toward fluent content remains. Social media research over the past decade documents that users readily share content without engaging with its accuracy, with inattention rather than belief driving much of the sharing behavior [34], and the operational reviewer encountering polished AI output sits inside a behavioral pattern that predates LLMs. Detection may remain possible where provenance metadata, watermarking, platform logs, or controlled generation environments are available, but these mechanisms do not solve the problem faced by ordinary reviewers encountering polished text in operational workflows. The governance question should therefore shift from "was this AI-generated?" to "was this meaningfully reviewed?"

The current state of practical AI observability illustrates this gap concretely. Mature AI observability stacks already combine model-trace tooling (Langfuse, Phoenix), annotation infrastructure for human review and correction data (Braintrust, Langfuse annotation queues), workflow process mining to verify that human review occurs in the right sequence (Celonis, ServiceNow), and governance platforms for ownership and policy mapping (Credo AI, Holistic AI).[1] Each layer addresses a real component of HITL or HOTL failure. None directly measures oversight quality, because human review behavior is rarely instrumented as part of the production control loop. The functional quality of human review remains uninstrumented in current production deployments, and the absence is not generally recognized as a measurement gap in the way that, for example, hallucination detection is.

The result is a compounding failure. Detection infrastructure that was designed to support oversight becomes, under LLM deployment conditions, a mechanism that reinforces the oversight degradation it was designed to prevent. Fluent outputs that clear detection thresholds arrive at reviewers pre-endorsed by the absence of alerts, which reduces the probability of substantive review precisely for the outputs that most require it.

4.2 Interpretability

The interpretability literature frames the central problem as making AI decision-making legible to human reviewers. Methods range from local explanation techniques that surface the features driving individual predictions [37] to mechanistic interpretability research that examines internal model representations. These methods share a common design assumption: that if the model's reasoning can be made visible, the human reviewer will engage with what has been made visible. The assumption is not unreasonable as a design premise. It does not hold as an operational guarantee.

Two findings from the reliance literature are directly relevant. Fok and Weld demonstrate that explanations improve human-AI complementarity only insofar as they enable the reviewer to verify whether the model's reasoning is correct for the specific case at hand; explanations that are plausible but not verifiable do not produce appropriate reliance and may increase confidence without increasing accuracy [12]. Schemmer and colleagues operationalize appropriate reliance as a measurable construct and find that explanation provision does not reliably produce it [41]. Interpretability methods can increase the legibility of model reasoning without increasing the functional quality of oversight, because legibility and engagement are not the same thing.

There is a further complication specific to LLM-mediated knowledge work. Explanations produced by or about language models are themselves linguistic artifacts. A reviewer evaluating a fluency-rich primary output and a fluency-rich explanation of that output faces the same suppression of scrutiny signals at both layers. The explanation does not interrupt the cognitive pattern that produces procedural confirmation; it provides additional content that the same pattern can confirm procedurally. Richer interpretability infrastructure can therefore coexist with, and may in some conditions reinforce, the degraded oversight state it was designed to prevent.

4.3 Governance

Governance frameworks for AI deployments are designed to maintain meaningful oversight through defined roles, approval chains, and documented processes. The NIST AI Risk Management Framework structures oversight around risk identification, measurement, and management, with guidance on accountability for AI deployment decisions [47]. Model cards and datasheets provide structured documentation of model capabilities, limitations, and intended use contexts, intended to support reviewer awareness of where outputs warrant additional scrutiny [14, 26]. Internal algorithmic auditing frameworks adapt established audit practices to AI deployments, creating end-to-end documentation of design choices and review decisions [35]. Internal audit frameworks more broadly adapt established control assurance methods to AI-specific risk categories, providing structured evidence that documented controls operated during the audit period. Each of these mechanisms is valuable, and

[1] Vendor documentation: Langfuse (https://langfuse.com), Phoenix by Arize (https://phoenix.arize.com), Braintrust (https://braintrust.dev), Celonis (https://www.celonis.com), Credo AI (https://www.credo.ai), Holistic AI (https://www.holisticai.com). Tools are cited as illustrative of the current state of practical AI observability; inclusion does not constitute endorsement.

each contributes evidence that the oversight role was defined, documented, and procedurally activated. None directly observes whether the cognitive function the role was designed to fulfill was actually exercised at the moments that matter. The limitation is partly historical: in the author's experience across academic and enterprise deployments, governance structures in organizations are often afterthoughts, bolted on in response to a lapse, incident, regulatory pressure, or existential threat rather than designed proactively for the systems they come to govern. The frameworks therefore instrument what their originating conditions made salient, primarily process compliance and documentation integrity, rather than what subsequent operational conditions require. The limitation is compounded by partial implementation: in operational settings the author has observed, large governance frameworks, including the AI RMF and adjacent frameworks like MITRE ATLAS, are typically deployed in fragments selected for tractability or auditor satisfaction rather than for completeness against the failure modes the frameworks were designed to address. The fragments most commonly implemented are the documentation and process-compliance components, which are easier to operationalize than the components that would require new instrumentation. This produces the limitation that maps onto the observational identity problem established in Section 3: organizations adopt the parts of governance frameworks that verify review occurred without adopting the parts that would observe whether the cognitive function those reviews are supposed to represent was actually exercised.

Green's analysis of 41 policies that prescribe human oversight of government algorithms finds that these policies rest on an unexamined assumption that humans are able to perform the oversight functions they are assigned: they mandate presence without verifying capacity or engagement [17]. The result is that oversight requirements can legitimate AI deployments without ensuring the oversight that legitimation implies. A governance framework that certifies process integrity without observing functional integrity provides a form of assurance that the degradation pattern systematically undermines.

This pattern is not confined to algorithmic governance research. In operational contexts the author has worked in, employees trained on regulatory and compliance procedures often abandon that training under operational pressure, returning to throughput-driven defaults the moment they are back in the high-pressure context the training was meant to govern. The presence of compliance training in the record does not constitute evidence that compliant judgment was operating at the moment of decision.

The governance gap has a specific structural dimension that critiques in the existing literature have not fully articulated. Governance frameworks were designed for conditions of intentional non-compliance: an employee who skips a required review step, a team that bypasses a mandated approval process, an organization that documents a process it does not follow. The detection mechanisms embedded in governance frameworks, audits, compliance checks, exception reporting, are calibrated for these conditions. The degradation pattern described in Section 3 is not non-compliance. Every step is followed. Every approval is recorded. Every required party participates. The governance framework reports full compliance because full compliance is occurring. What is absent is the functional quality that compliance was designed to ensure, and governance infrastructure has no sensor for that absence.

Convergent work from computational reduction theory reaches a structurally adjacent conclusion through a different formal route. Chiodo and colleagues formalize HITL setups using oracle machines from computability theory, distinguishing trivial monitoring, single endpoint human action, and highly involved human-AI interaction, and derive a taxonomy of HITL failure modes that highlights the limitations on what any HITL setup can actually achieve regardless of how it is procedurally specified [7]. Their analysis identifies an unavoidable trade-off between the attribution of legal responsibility and the technical explainability of AI, and argues that the choice of HITL setup carries technical and ethical implications that current legal frameworks do not adequately recognize.

A related strand of work arrives at the governance-record problem from the supervision side rather than the computability side. Manheim and Homewood distinguish oversight (ex-post, policy and governance) from control (ex-ante or real-time, operational) and argue that the requirements for "human oversight" in regulatory texts including the EU AI Act risk codifying vague or inconsistent interpretations of supervision, with the consequence that claims of oversight not backed by inspectable documentation become difficult to distinguish from performative safety presentation [25]. Their oversight-versus-control distinction is complementary to the supervisory-control framing used here, and their argument that undocumented oversight claims approach safety theater independently supports the central concern this paper develops: that procedural records of compliance cannot themselves substantiate the functional engagement they purport to evidence.

The framework developed in this paper approaches the same problem space from observational and supervisory-control directions and reaches a complementary conclusion: HITL setups that satisfy procedural and legal requirements can fail to deliver the operational engagement those requirements were designed to ensure, and existing governance infrastructure has no mechanism to observe the difference.

4.4 The Compounding Structure

The three gaps do not operate independently. They compound. The distinction between model-centric and oversight-centric transparency makes this visible. Table 1 maps the two orientations across four dimensions. These orientations are complementary rather than competing: model-centric transparency addresses real problems in AI deployment. The argument here is that it is insufficient, because the oversight layer has not been treated as a component requiring its own instrumentation.

Detection infrastructure pre-endorses outputs that clear fluency thresholds, reducing the probability of substantive review. Interpretability infrastructure provides additional legible content that the same cognitive pattern can confirm procedurally. Governance infrastructure records the resulting approvals as compliant review events. Each layer of the transparency stack performs its designed function while contributing to a system state in which governance records certify oversight that has silently ceased to be substantive.

Table 1: Model-Centric Versus Oversight-Centric Transparency

Dimension	Model-Centric Transparency	Oversight-Centric Transparency
Primary object of instrumentation	Model reasoning, outputs, and behavior	Functional state of the oversight mechanism
Evidence produced	Interpretability maps, audit logs, governance artifacts	Engagement metrics, correction-rate trajectories, calibration indicators
Failure mode addressed	Hallucination, bias, distribution shift	Authority drift, procedural confirmation, oversight formality
Human role assumed	Stable recipient of explanations	Degradable system component requiring care and feeding

5 Toward Oversight-Aware Transparency Research

The analysis in Section 4 identifies a shared limitation across detection, interpretability, and governance methods: each treats the oversight function as a stable substrate rather than as a system component requiring care and feeding. Closing this gap requires extending the scope of transparency and control research to include the oversight layer itself. This section proposes three research directions organized around instrumentation, structural conditions, and architectural design.

5.1 Instrumenting the Oversight Layer

The first research direction is to develop oversight-layer instrumentation that is integrated with the model-layer transparency stack rather than added as a separate compliance documentation process. Existing approaches to instrumenting oversight largely take the form of post-hoc activity logs, compliance trackers, and intervention counters operating in parallel with rather than as part of model observability. The research opportunity is to bring oversight-quality indicators into the same data substrate as model-trace tooling, so that structural signals of oversight degradation can be reasoned about alongside the model-layer signals they are meant to backstop. The central challenge is methodological: the functional state of the reviewer is not directly observable, and approaches that require individual behavioral tracking raise legitimate privacy concerns, governance concerns, and concerns about systematic bias in human evaluation, including familiarity effects, variability in subject-matter expertise, recency bias, implicit bias, and other distortions that make individual-level engagement signals unreliable even where the technical capacity to collect them exists. The instrumentation agenda should therefore prioritize structural and aggregate indicators that reflect oversight quality at the workflow or deployment level without requiring surveillance of individual reviewers.

The structural logic is familiar from incident management, where signals of attention degradation (alert volume relative to investigation depth, mean time to acknowledge versus mean time to resolve, escalation completion rates) are routinely treated as system health indicators rather than as performance evaluations of individual responders. The candidate indicators below transpose that logic to the oversight quality of AI-assisted workflows, drawing on established lines of human factors and AI reliance research including override rates, validation load in alert fatigue contexts, and workflow dependency analysis. They are introduced as hypotheses and design proposals, not as validated metrics; their interpretation depends on confounds discussed throughout this section and on empirical work that the present paper opens but does not complete. Table 2 summarizes each indicator, its measurement approach, what it captures, and its primary interpretive limitation.

Table 2: Candidate Indicators for Oversight Layer Instrumentation

Indicator	Measurement Approach	What It Captures	Primary Limitation
Correction trajectory	Rate of modification, rejection, or escalation of AI outputs over deployment duration	Aggregate reviewer engagement relative to output volume	Confounded by system maturity; requires discrimination protocol
Validation load	AI outputs requiring review per unit of available reviewer capacity and time	Structural conditions under which oversight operates	Does not directly measure engagement quality; measures risk conditions
Dependency accumulation	Number of downstream workflow nodes dependent on outputs passing a given checkpoint	Consequence visibility at point of decision	Requires workflow graph instrumentation; static in some architectures

As a concrete illustration of how these indicators operate in practice, consider correction-rate trajectory. A team deploying an LLM-assisted document review workflow logs that 30% of AI-generated drafts received substantive corrections from human reviewers in Week 1 of deployment. By Week 4, the rate has dropped to 10%, with no documented changes to the model, prompts, or output type distribution, and no measurable improvement in independent quality metrics. The decline is consistent with oversight drift rather than system maturity, and the discrimination protocol developed below provides the structural signals required to differentiate the two.

Each indicator requires interpretive discipline. The most significant confound is the distinction between oversight degradation and system maturity. Both produce declining correction rates over deployment duration, but through different mechanisms. System maturity produces correction-rate decline through genuine output quality improvement. Oversight degradation produces correction-rate decline through reduced reviewer engagement: the threshold for intervention rises while output quality remains stable or variable. Discriminating between these mechanisms requires attending to three signals: correction-rate decline in the absence of model updates or measurable quality improvement; divergence between declining correction rates and stable or increasing downstream error rates; and uniform correction-rate decline across output types including those that have historically produced errors.

A related concern is the potential conflation of oversight quality with productivity or workflow efficiency metrics. Declining correction rates, faster approval times, and reduced escalation frequency are consistent with both improved workflow efficiency and degraded oversight quality. Governance frameworks that treat these metrics as positive indicators may be systematically misreading oversight degradation as operational improvement. Downstream error rate conditioned on AI confidence level, disagreement rate between reviewers and AI recommendations under controlled conditions, and calibration stability across personnel transitions are candidate metrics that capture oversight quality without conflating it with efficiency. Research on developer productivity measurement provides relevant methodological grounding for constructing multi-dimensional quality indicators that resist single-metric gaming [13].

A further design constraint applies to the instrumentation agenda: the indicators themselves must be resistant to the substitution dynamic they are designed to detect. Perverse incentives in measurement are a familiar problem to anyone who has watched a team optimize for jira ticket close rate rather than feature delivery, for token usage rather than user value delivered, or for severity-zero incidents resolved rather than incidents prevented from happening. Goodhart's Law generalizes the pattern: any observed statistical regularity collapses once pressure is placed upon it for control purposes [15], a principle commonly rendered in the formulation that when a measure becomes a target, it ceases to be a good measure [46]. Performance metrics that begin as proxies for valued outcomes routinely come to be optimized directly, displacing the outcomes they were meant to track. If correction rate is monitored as a proxy for engagement quality and reviewers are aware of that monitoring, the rational response is to generate corrections that satisfy the metric rather than corrections that improve output quality. The surrogation mechanism identified by Choi and colleagues [8] applies with equal force to governance instrumentation as to the oversight workflows it is designed to observe: once correction rate becomes the measured attribute, it will substitute for engagement quality in the very cognitive processes that instrumentation was intended to surface. The general design principle is that oversight quality indicators should be chosen such that satisfying the metric and exercising genuine oversight are difficult to decouple.

Several specific instrumentation approaches warrant empirical testing. Each is described in terms of what a team would build, what data the system would emit, and what assumption the indicator rests on. None has been validated at the level required for governance certification, and each is a candidate solution to be refined under deployment conditions.

Seeded calibration checks. Inject a small fraction of pre-validated outputs into the review queue, mixed indistinguishably with normal traffic. The validation source can be a domain expert panel, a separate model run with known reliability, or held-out ground-truth cases. Log the reviewer's disposition on these seeded items and compute disagreement rate against the known answer. A reviewer engaged with the work disagrees on seeded errors at roughly the rate the validation source identified them; a reviewer in procedural mode approves seeded errors at the rate the model produced them. Implementation requires a sampling rate, a validation pipeline, and routing logic that prevents reviewers from identifying seeded items. The assumption being tested is that disagreement rate on seeded items tracks engagement quality on non-seeded items.

Onboarding-period baseline checks. When a new reviewer joins the workflow, log their correction rate, modification rate, and disagreement pattern for a defined onboarding window before they are calibrated to operational norms. Compare against the prevailing baseline of incumbent reviewers. A persistent gap, where new reviewers correct more or disagree more than incumbents, signals that the operational baseline has drifted from the Phase 1 calibration that documented processes still describe. The system requires defined onboarding windows, stable role definitions across hires, and access to incumbent baseline data. The assumption being tested is that fresh reviewers operating against ground truth produce a more reliable engagement signal than long-tenured reviewers operating against accumulated norms.

Dependency surfacing at decision time. At each review checkpoint, compute the number of downstream workflow nodes that depend on the output passing the checkpoint. Surface this count to the reviewer in the interface, with visible consequence information for high-dependency cases. Log review duration, modification rate, and approval rate as a function of dependency weight. If engagement scales with surfaced dependency, the intervention is working; if engagement remains flat across dependency levels, the surfacing is not producing the intended effect and the design needs revision. This requires workflow graph instrumentation and an interface change. The assumption being tested is that legible consequence information sustains substantive review under load.

Decision-point logging for long-running workflows. Agentic systems that produce a single consequential output after extended autonomous operation require trajectory-level instrumentation, not just output-level review. Build the agentic system to emit structured events at each decision point that meets defined criteria: irreversibility, resource expenditure above a threshold, external action, or state changes that affect other systems. Route these events to a reviewer with the context required to evaluate the proposed action before it executes. Log intervention rate at flagged decision points, latency between event emission and reviewer response, and the proportion of trajectory operations that proceed without any human contact. The system requires the agent to expose its decision structure rather than only its final output, which is a design constraint that not all current agentic architectures satisfy. The assumption being tested is that decision-point review produces better oversight outcomes than artifact-level review for long-running workflows.

Workload ceilings with backpressure. Validation-load ratios above a measurable threshold predict the shift to fast, automatic pattern recognition becoming dominant regardless of reviewer training or motivation. Implement explicit ceilings on outputs-per-reviewer-per-unit-time based on task complexity, with the routing system refusing to assign additional outputs once a reviewer hits ceiling. Outputs queue or route to other reviewers rather than entering a saturated review state. Log the rate at which ceiling backpressure triggers, the downstream error rate before and after ceiling enforcement, and reviewer-reported workload pressure. This converts an observational indicator into an architectural enforcement mechanism. The assumption being tested is that ceiling-enforced workflows produce measurably better oversight outcomes than uncapped workflows under matched output volume and task complexity.

A complementary direction is incident attribution instrumentation. When an AI-assisted workflow produces a consequential failure, post-incident root cause analysis requires reconstructing the oversight state at the relevant decision points, not just the technical chain of causation. This requires structured event emission at all AI-involved checkpoints capturing output provenance, reviewer disposition, review duration, active alerts, and concurrent oversight-quality indicator values, retained for a window sufficient to span incident-discovery latency. Without this instrumentation, incidents traceable to procedurally confirmed AI outputs are indistinguishable in the post-incident record from incidents traceable to substantively reviewed AI outputs. The observational identity problem applies to incident attribution as much as to ongoing oversight.

Each of these can be built with current workflow infrastructure and existing logging capabilities. None requires research-grade measurement instrumentation. They are candidate solutions to be deployed, observed, and refined rather than validated indicators ready for governance certification. The general design principle remains that satisfying the metric and exercising genuine oversight should be difficult to decouple. The approaches above are practical operationalizations of that principle, positioned at the intersection of research inquiry and engineering practice, where the next round of empirical work needs to happen.

An empirical program currently underway examines, as its initial study, whether time pressure measurably reduces oversight quality in LLM-assisted knowledge work using behavioral indicators of engagement rather than approval records alone, with a parallel investigation of whether topology-aware enforcement of review requirements reduces downstream error propagation relative to no-enforcement baselines. These studies represent the beginning of a broader program; establishing validated oversight quality metrics deployable in operational governance settings will require multi-site replication, longitudinal study designs, and engagement with the domain-specific moderators identified in Section 6.1.

5.2 Characterizing Structural Conditions

The second research direction is to characterize the structural conditions that accelerate oversight degradation. The conditions fall into three categories: temporal, organizational, and incentive-structural.

Temporal conditions. These concern the relationship between deployment duration and calibration state. Reviewers entering a deployment develop a working sense of what acceptable output looks like, calibrated against the outputs they encounter early in the deployment. Without periodic recalibration, defined as deliberate exposure to known-correct and known-incorrect outputs against which the reviewer's judgment can be retuned, that initial calibration becomes the operational baseline. Deviation from the baseline is then required to trigger evaluation, and the baseline itself drifts over time toward whatever the model is currently producing. Personnel turnover that breaks calibration without breaking process is a specific and consequential variant: incoming reviewers inherit documented processes that reflect Phase 1 design intent but operational norms that reflect Phase 3 reality.

Organizational conditions. These concern the structural relationship between output volume, reviewer capacity, and task complexity. When output volume relative to validation capacity exceeds what careful review can accommodate, procedural confirmation is a rational response to unsustainable cognitive load. Under these conditions, deliberate analytical engagement with each output is structurally unavailable, and pattern recognition against expected output structure becomes the dominant processing mode. Parallel model proliferation without corresponding validation infrastructure is a related condition: as the number of AI systems requiring review increases, validator calibration to any single system's failure signatures degrades. Self-serving incentive structures compound this further: when individuals are rewarded for their own output rather than collective validation quality, conformity to expected patterns increases and diversity of scrutiny declines, degrading the collective intelligence of the oversight function [3].

Incentive-structural conditions. These represent the category requiring most development. When organizational incentives reward throughput, output volume, or process completion rather than validation quality, they create systematic pressure toward procedural confirmation that operates independently of individual motivation, training, or intent [18, 20]. A reviewer rewarded for speed of approval is not making a cognitive error when oversight degrades; they are responding rationally to the reward structure they face [23]. Oversight frameworks that do not account for incentive alignment cannot reliably sustain substantive oversight regardless of how clearly they mandate it: the mandate and the incentive must be aligned, or the mandate will be satisfied procedurally while the incentive shapes actual behavior [17].

The cognitive mechanism connecting incentive structure to oversight degradation operates through attribute substitution and motivated reasoning [19, 21]. When throughput is measured and validation depth is not, process completion substitutes for substantive evaluation in practitioners' cognition. This substitution is not deliberate; it is the non-conscious product of an incentive architecture that makes one attribute salient and the other invisible [8]. The motivated reasoning mechanism has substantial empirical support in the adjacent literature on incentive-induced unethical behavior: a synthesis of 361 studies finds that goal-driven motivated reasoning, in which an incentivized outcome comes to displace other evaluative considerations in practitioner cognition, is among the more empirically established mechanisms linking incentive structures to unethical behavior in education, health care, and business settings, though most of the supporting evidence is drawn from laboratory experiments rather than field studies [33]. The result is that practitioners can simultaneously believe they are performing their oversight function and be functionally failing to do so.

Incentive structures that create time pressure directly accelerate the temporal conditions above, compressing the window available for deliberate analytical engagement. Incentive structures that reward output volume without quality gates remove the consequence that would otherwise make procedural confirmation costly [42]. Incentive structures that evaluate reviewers on throughput while holding them nominally accountable for errors create an asymmetric risk environment in which the cost of slow review is immediate and visible while the cost of degraded review is deferred and diffuse.

A further dimension concerns multi-dimensional attribution. In human-AI oversight contexts, reviewers receive feedback in environments where both their own judgment and the model's output contribute to outcomes. Coutts, Gerhards, and Murad show that self-serving attribution bias is specifically amplified under these conditions: when feedback involves another agent's contribution, individuals distort their belief updating in self-serving directions in ways that do not occur when the other factor is random [10]. A reviewer who approves an output that later proves incorrect may attribute the error to the model rather than to their own failure to scrutinize, sustaining an inflated self-assessment of oversight quality that accelerates Phase 2 transition.

Early observational patterns suggest that some communities and deployment contexts exhibit responses more severe than procedural confirmation alone, including patterns consistent with the cognitive surrender phenomenon documented by Shaw and Nave [43], in which evaluative judgment shifts from the person to the model rather than being supplemented by it. The conditions under which professional identity norms either resist or accelerate these responses represent a convergence point between the incentive-structural literature and research on how organizational and community contexts shape implicit cognitive responses to external pressure [9]. A fuller empirical treatment of incentive-structural conditions in AI deployment contexts, drawing on principal-agent theory [18] and the performance metric dysfunction literature [20], is a priority direction for the research program this paper opens.

5.3 Designing for Degraded-State Conditions

The third research direction reframes the design problem for oversight architectures. Current design practice optimizes oversight mechanisms for nominal conditions: sufficient reviewer capacity, adequate time, appropriate training, functioning incentives. The degradation pattern indicates that these conditions are not stable. A more robust design approach treats degraded-state behavior as an explicit design requirement rather than an edge case.

Observational work on LLM-assisted workflows in academic and enterprise settings has identified two design principles relevant to oversight architecture under degraded conditions [27]. The first is that validation conditions should be enforced structurally rather than assumed procedurally. An architecture in which the system enforces review requirements, for example by requiring evidence of substantive engagement before approval is recorded or by making the downstream consequence of approval visible at the point of decision, is less vulnerable to degradation than one that relies on reviewer compliance under operational pressure. The second principle is that oversight architecture should be designed with explicit attention to its failure modes under sustained operational conditions. The implementation approaches developed in Section 5.1, including workload ceilings, dependency surfacing, and seeded calibration checks, are concrete instances of structural enforcement applied to specific conditions.

These principles suggest a design agenda that connects oversight instrumentation to oversight enforcement. If structural indicators can detect early-phase degradation, those indicators can trigger architectural responses, recalibration events, load redistribution, or escalation requirements, before the degradation reaches the terminal phase.

5.4 Implications for Governance Frameworks

Frameworks that certify AI deployments on the basis of process compliance without instrumenting oversight quality provide weaker safety guarantees than their documentation implies. Strengthening these frameworks does not require abandoning existing mechanisms; it requires adding an instrumentation layer that current frameworks lack.

Concretely, governance frameworks should specify not only that human review occurs but the conditions under which review can be substantive, including reviewer capacity, validation load constraints, calibration refresh requirements, and structural indicators that trigger review of the review process itself. Audit mechanisms should distinguish between process integrity and functional integrity, recognizing that a procedurally complete record is necessary but not sufficient evidence of effective human control. Governance frameworks should also examine whether the local incentive environment, including throughput targets and performance evaluation criteria, is consistent with the oversight conditions the framework mandates, since incentive misalignment can render mandated review procedural even where the framework is fully complied with. These are analytical recommendations; their implementation in specific organizational and regulatory contexts requires empirical validation and context-specific judgment that the present paper does not supply.

6 Limitations and Future Directions

The argument developed in this paper is analytical and conceptual. Its claims are grounded in observational evidence and established literature, but they are not validated through controlled experiment, and the instrumentation directions proposed in Section 5 have not been empirically evaluated in deployed organizational settings.

6.1 Scope of Observational Evidence

The degradation trajectory described in Section 3 is drawn from two deployment contexts, one academic and one enterprise. Both are knowledge-work settings involving LLM-assisted document and artifact production. The generalizability of the three-phase pattern to other domains, including clinical decision support, legal review, financial analysis, and public-sector algorithmic governance, is plausible given the structural similarities in the oversight conditions involved, but has not been established empirically. Domain-specific factors, including professional training, regulatory environment, task reversibility, and accountability structure, may modulate the rate and character of degradation in ways the present framework does not yet capture.

The cross-domain analogue literature provides partial support for generalizability. Alert fatigue in clinical decision support systems, where override rates increase with task complexity and repeated exposure [1], monitoring degradation in aviation, where trained pilots lose situational awareness of automation state under conditions that make intervention most necessary [40], and alert fatigue in security operations centers, where analysts become desensitized to high alert volumes with documented consequences for threat detection [48], all describe structural patterns consistent with the degradation trajectory identified here. These analogues suggest that the pattern is not idiosyncratic to LLM-assisted knowledge work. They do not constitute direct evidence that the three-phase structure, the observational identity problem, or the governance gap manifest in the same form across these domains.

6.2 The Measurement Problem

The candidate indicators proposed, correction-rate trajectory, validation-load metrics, and dependency accumulation patterns, are structural proxies for oversight quality rather than direct measures of it. The most significant confound is the distinction between oversight degradation and system maturity, both of which produce declining correction rates through different mechanisms. A governance framework that treats declining correction rates as a positive signal without this discrimination is vulnerable to misreading oversight degradation as operational improvement.

A related confound is the conflation of oversight quality with workflow efficiency. Oversight quality metrics must be analytically independent of throughput measures. Downstream error rate conditioned on AI confidence level, disagreement rate between human reviewers and AI recommendations under controlled sampling conditions, and calibration stability across personnel transitions are candidate metrics that partially satisfy this requirement. Their full validation requires controlled study designs that are not yet available in operational governance contexts.

6.3 The Surveillance Boundary

The instrumentation agenda raises a normative concern that the paper does not fully resolve. The candidate indicators proposed in Section 5 are aggregate and structural, designed to avoid individual behavioral tracking. However, the line between system-level instrumentation and individual monitoring is not fixed; it depends on implementation choices, data retention policies, and the granularity at which indicators are computed and reported.

Work on effective oversight frames this concern as a design problem rather than a categorical prohibition [45]. Effective oversight depends not only on individual reviewer capacity but on technical design and environmental conditions that either support or undermine substantive engagement. Instrumentation aimed at those conditions is meaningfully different from instrumentation aimed at individual reviewer behavior.

A workable design principle for distinguishing oversight-layer instrumentation from individual surveillance is that the indicators must be structural and aggregate by construction, not by policy. Three requirements follow. First, the data architecture itself should make individual-level access infeasible at the granularity that would permit performance monitoring of named reviewers, rather than relying on policy commitments not to use available data in that way. Second, the indicators should serve the system-level oversight function alone; exposure to management evaluation, productivity scoring, or compensation systems converts the instrumentation into the surveillance condition it was designed to avoid. Third, threshold conditions that trigger backpressure or remediation should operate on aggregate workflow state rather than on individual reviewer state, so that intervention takes the form of system adjustment rather than personnel management. These requirements are articulable independently of specific governance contexts, even as their operationalization requires engagement with privacy frameworks, labor relations considerations, and organizational accountability structures that are outside the scope of the present paper.

6.4 Future Directions

The limitations above define not a ceiling on this research but a foundation for it. The instrumentation agenda, the structural conditions characterization, and the architectural design principles proposed in this paper each represent multi-year research threads whose full development requires methodological diversity, multi-site replication, and engagement with domain-specific moderators that a single study or deployment context cannot provide.

The immediate empirical agenda addresses three questions. First, whether the degradation trajectory described in Section 3 replicates under controlled conditions and across deployment domains, with explicit attention to discriminating maturity-driven from engagement-driven correction-rate decline. Second, whether the candidate structural indicators proposed in Section 5.1 track oversight quality in operational settings, and whether governance frameworks built around those indicators produce measurable improvements in downstream error detection relative to process-compliance baselines. Third, whether oversight architectures designed for degraded-state conditions reduce Phase 3 outcomes without imposing usability costs that make substantive review unsustainable under normal operational conditions.

These questions open onto a broader set requiring more sustained investment. How do incentive structures interact with architectural design to produce different degradation trajectories across organizational contexts? What are the domain-specific moderators that determine whether the three-phase pattern manifests in clinical, legal, financial, and public-sector AI deployments? How should oversight quality metrics be integrated into regulatory certification frameworks? What community-level and cultural conditions make some deployment populations more vulnerable to pathological responses including cognitive surrender and intellectual replacement of human judgment?

Each of these questions represents a research thread that the present paper opens but does not close. The uninstrumented oversight layer is not a problem that a single measurement instrument or architectural intervention will resolve. It is a structural property of how human-AI systems have been built and governed at scale, and addressing it requires a research program commensurate with that scale.

The empirical phase of that program is underway at the Human Oversight Systems Lab. Its near-term focus is the discrimination protocol between maturity-driven and engagement-driven correction-rate decline in controlled deployment conditions, and the structural validation of candidate indicators against established measures of attentional engagement. The program would benefit from three kinds of partnership: deployment-context partners willing to instrument oversight quality in operational settings under the structural-and-aggregate constraints articulated in Section 6.3; regulatory and standards partners working on how oversight requirements might be made functionally rather than procedurally auditable; and methodological partners with expertise in the cognitive dynamics of evaluative judgment under fluent-text review conditions.

LLM Usage Disclosure

Tools and scope. This paper was developed through sustained collaboration with Claude (Anthropic) across literature identification, argument development, manuscript drafting, and response to reviewer feedback. The author maintained supervisory control over research questions, observational evidence, theoretical commitments, interpretation, and all conclusions, and takes full responsibility for every claim in this submission.

Nature of the contribution. Claude's involvement in the manuscript took the form of substantive edits: drafted passages that the author reviewed, restructured, accepted in part, rejected in part, or replaced with original prose. The author treated each substantive edit as a candidate that required independent verification of its content, sourcing, and epistemic register before incorporation. The acceptance rate for unmodified passages was not high. The acceptance rate for passages incorporated after revision was higher. The boundary between AI-assisted drafting and original authorship is not separable into discrete contributions; what is reportable is the supervisory architecture under which substantive edits were evaluated and the failure modes that recurred across that evaluation.

Observed failure modes. Four failure modes recurred consistently across the manuscript development process and are reported here as practitioner observations rather than empirical findings. The specific signatures observed (overused vocabulary, formulaic sentence and conclusion structures, voice-flattening tendencies) are consistent with the AI-generated-text signatures independently catalogued by expert annotators in recent empirical work [38], providing convergent evidence that the failure modes described below are detectable patterns rather than idiosyncratic observations.

Hallucination under instruction imprecision. Output quality is sensitive to prompt specificity. Imprecise instructions produced fabricated citations with plausible formatting, invented publication details, and misattributed claims. Active verification against primary sources was required before any citation was incorporated. The verification discipline was a designed component of the workflow rather than an incidental check, and failures caught in verification were frequent enough to establish that passive acceptance of model-generated references would have introduced material errors into the published record.

Overclaiming as the dominant failure mode. Across the manuscript development process, overclaiming was the most frequently encountered failure mode. Outputs consistently tended toward stronger assertions than the evidence supported, broader generalizations than the observational base warranted, and more confident causal claims than the analytical structure justified. Consistent downward calibration was required throughout. This pattern is consistent with the fluency suppression argument developed in Section 2: outputs that are internally coherent and well-formed suppress the scrutiny signals that would ordinarily trigger correction, which makes overclaiming particularly difficult to catch without deliberate adversarial review.

Voice drift requiring active validation. Sustained LLM collaboration produces a gradual flattening of the author's established epistemic register. Outputs tend toward a generic academic prose style that is competent but epistemically distinct from the author's prior published work. Without active correction, this produces a manuscript that reads as written by a capable but anonymous academic rather than by a researcher with a specific intellectual history and argumentative stance. Active validation against prior published papers was required throughout the development process, and a voice validation workflow was developed to systematize this check. The development of that workflow is itself a finding: voice maintenance is not automatic and cannot be assumed.

Substantive engagement as effortful and degradable. The most consequential observation is that substantive engagement with model outputs is not a stable condition. It is recoverable but effortful, and its maintenance requires deliberate architectural choices about session structure, review cadence, and what is treated as requiring independent verification versus procedural confirmation. Under conditions of high output volume, time pressure, or extended session duration, the author observed in their own practice the transition from active evaluation to pattern recognition that Section 3 describes. This is not a claim about a controlled experiment; it is a candid report of a practitioner's experience with the degradation pattern in the context of their own research production.

Reflexive position. The development of this paper became an instance of the phenomenon under study. The four observations above are consistent with the degradation pattern the paper describes and with the broader reflexive literature on AI-assisted research methodology [49]. The author offers them not as validated findings but as observational evidence consistent with the analytical argument the paper advances.

Disclosure context. Recent analysis indicates that LLM tools are involved in a substantial and rapidly growing share of scholarly publishing, while visible disclosure remains rare and uneven [16]. The central integrity problem is not LLM use, but unattributed LLM use: a governance record that certifies a paper was written by human authors cannot distinguish substantive human authorship from procedural confirmation of model-generated text. This is structurally identical to the observational identity problem the paper describes in the context of AI-assisted knowledge work. Full and honest disclosure does not resolve this problem at the field level, but it is one instance of the oversight-centric transparency the paper argues for: making the functional state of human engagement with model outputs visible rather than assuming it from the procedural record of authorship.

Responsibility. The author takes full responsibility for every claim in this submission.

Acknowledgments

The author thanks the AITC 2026 reviewers for their close reading and substantive comments, and Maurice Chiodo (Centre for the Study of Existential Risk, Cambridge) for the pointer to the Schufa-slide regulatory finding that sharpened the framing in Section 4.3.

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